INCH-POUND

MIL-T-27/99B
13 March 1991
SUPERSEDING
MIL-T-27/99A
15 NOVEMBER 1984

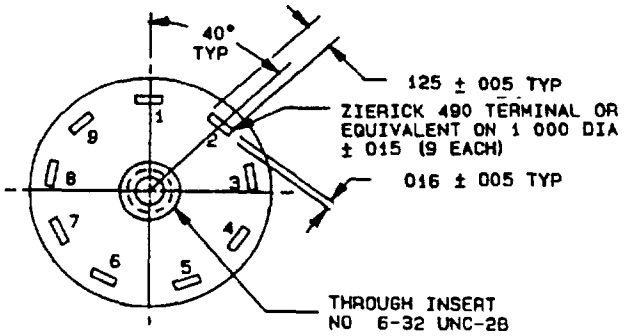
MILITARY SPECIFICATION SHEET

TRANSFORMER, POWER, STEP-DOWN, TF5S03ZZ

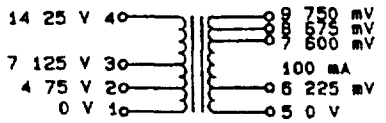
Ⓑ Inactive for new design after the date of this specification.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-T-27.

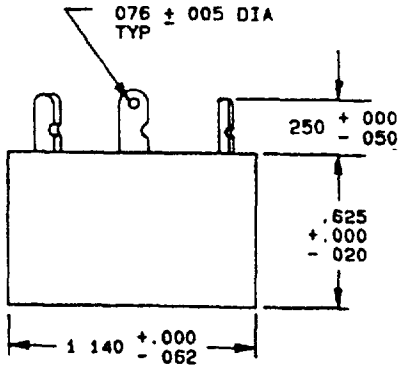


CIRCUIT DIAGRAM AND MARKING



VA 075 V

WORKING VOLTAGE 30 V MAX



Inches	mm	Inches	mm
.000	0.00	.062	1.57
.005	0.13	.076	1.93
.015	0.38	.125	3.18
.016	0.41	.250	6.35
.020	0.51	.625	15.88
.050	1.27	1.140	28.96

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Tolerance on terminal angular position is ±0.5°.
4. Marking shall be on the side of the case.

FIGURE 1. Dimensions and configuration.

Ⓑ denotes changes

DISTRIBUTION STATEMENT A. Approved for public release, distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T03:31Z
Check the source to verify that this is the current version before use.

REQUIREMENTS: (When numbers in parentheses, i.e., (1-2) are used, they indicate the winding and the extreme terminals of the winding.)

Electrical ratings.

Primary voltage (1-2-3-4): 4.75/7.125/14.25 volts pk, 3 kHz square wave.

Primary current.

(1-2). 16 mA.
(1-4). 8 mA.

Secondary voltage: (5-6-7-8-9): 225/600/675/750 mV pk.

Secondary current: (5-9): 100 mA.

Working voltage. (1-4) (5-9) 30 volts maximum.

Design and construction.

Dimensions and configuration See figure 1.

Duty cycle: Continuous.

Case. Encapsulated

Material. Epoxy.

Terminals. Brass, solder type.

Terminal height. 0.250 +.000, -.050 inch (6.35 +0.00, -1.27 mm).

Weight: 34.02 grams.

Operating temperature range: -40°C to +130°C.

Terminal strength MIL-STD-202, method 211, test condition A, 2.0 pounds

Dielectric withstanding voltage (at sea level). 500 V rms.

Electrical characteristics

No load. With 14.25 volts pk, 3 kHz across (1-4), the voltage across (5-6) shall be 0.25 volts pk ±3.0 percent, voltage across (5-7) shall be 0.617 volts pk ±3.0 percent, voltage across (5-8) shall be 0.695 volts pk 3.0 percent, and voltage across (5-9) shall be 0.77 volts pk ±3.0 percent.

Rated load: With 14.25 volts pk, 3 kHz across (1-4), the voltage across (5-9) shall be 750 mV pk ±10 percent.

DC resistance:

(1-4) 7.9 to 10.7 ohms.
(5-9): .040 to 060 ohm

Vibration, high frequency. MIL-STD-202, method 204, test condition A.

Shock (specified pulse) MIL-STD-202, method 213, test condition H.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T03:31Z
Check the source to verify that this is the current version before use.

Ⓑ Quality assurance provisions:

Qualification inspection. Not applicable.

Quality conformance inspection. Groups A and B tests of MIL-T-27 shall be applicable.

Marking location See figure 1.

Part number. M27/99-01.

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - MI
Navy - OS, SH
Air Force - 17, 80
DLA - ES

User activities:
Army - AR, ME
Navy - AS, MC
Air Force - 19

Preparing activity.
Army - ER

Agent:
DLA - ES

(Project 5950-0753-30)

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T03:31Z
Check the source to verify that this is the current version before use.